

Memo 03: MP1 Mass-Conservation Check and Updated Diagnosis

FVCOM-MP Delaware Test Cases

April 29, 2026

Purpose

This memo updates the MP1 mass-conservation diagnosis after rerunning case **a** with `SEDIMENT_MODEL=T`. The central comparison is now: case **a** as no-floc plastic with sediment runtime on, versus case **b** as floc-enabled but effectively null-kinetic/split-disaggregated plastic.

Archived Analysis Outputs

Two mass-conservation snapshots are now preserved under `FVCOM_MP_test_run/PYTHON/output/mass_conservation/archive/`.

Version	Archive label	Meaning
v001	v001_before_case_a_sediment_on	old comparison with case a using <code>SEDIMENT_MODEL=F</code>
v002	v002_case_a_sediment_on_full10d	new comparison with case a using <code>SEDIMENT_MODEL=T</code>

The live analysis folder has also been regenerated from the v002 state. The archived files include `mass_summary.csv`, `mass_timeseries_long.csv`, and all current comparison plots.

Current Evidence

All three current MAT files cover the same FVCOM model-day window, 58485.0–58494.957031, or comparison day 0.00–9.96.

Case	Sediment runtime	Final water (kg)	Final bed (kg)	Final total (kg)	Retained vs. a (%)
a	on	582.41	0.00	582.41	100.0
b	on	371.85	0.00	371.85	63.8
c	on	504.26	24.29	528.55	90.8

The major new finding is that enabling sediment runtime in case **a** reduced the final domain MP1 mass from the old archived value 1001.25 kg to 582.41 kg. Therefore a large part of the old case-**a**/case-**b** gap was not caused by the split floc path alone; it came from comparing different sediment-runtime states.

The remaining null-floc mismatch is still meaningful. With sediment runtime matched, case **b** ends 210.56 kg below case **a**, retaining 63.8% of the sediment-on no-floc baseline. Case **c** ends 53.85 kg below case **a**, retaining 90.8%, with 24.29 kg stored in the bed.

New Negative Test: BBL-Off Case **a**

To test whether sediment affected plastic only through sediment-controlled bottom roughness, a case-**a** diagnostic sediment input was prepared: `generic_sediment_a_no_bbl.inp`. This keeps `SEDIMENT_MODEL=T` and suspended sediment transport enabled, but sets `MB_BBL_USE=F`, `SG_BBL_USE=F`, `SSW_BBL_USE=F`, and `SSW_CALC_ZNOT=F`.

The preliminary run showed almost no change relative to the sediment-on case-**a** run with the original BBL settings. This is important negative evidence: the sediment/plastic interference is unlikely to be explained only by the Sherwood/Signell/Warner BBL roughness feedback. It also means that either the HPC build did not actively use that BBL path, or another sediment-enabled code path dominates the plastic difference.

Cap, Bed, and Split Checks

The hard concentration cap remains inactive in the present outputs. The maximum extracted concentrations are far below the code cap of 100 kg m^{-3} :

$$\max C_a = 2.41 \times 10^{-4}, \quad \max C_b = 6.98 \times 10^{-3}, \quad \max C_c = 1.45 \times 10^{-2} \text{ kg m}^{-3}.$$

The cap-threshold counters are zero for all cases with available counters. Thus `mod_plast.F:1242--1243` is still a non-conservative code site, but it is not active in this 10-day test.

For the null-floc comparison, bed exchange is also not the leading cause. Cases **a** and **b** both end with zero sampled bed MP1 mass and zero sampled deposition/erosion exchange. Case **c** does exchange with the bed, so it should be interpreted separately.

The saved floc split fields are internally consistent:

$$\text{mp1_agg} + \text{mp1_dis} \approx \text{mp1}.$$

The maximum total split-minus-combined mismatch is zero for case **b** and $1.83 \times 10^{-7} \text{ kg}$ for case **c**. Saved-output recombination is therefore not the present sink.

Sediment/Plastic Call Trace

The setup and timestep order make the sediment-runtime finding plausible.

1. **Setup order.** In `fvcom.F:643--651`, `SETUP_SED` is called before `SETUP_PLAST`. Thus the sediment module is initialized first when `SEDIMENT_MODEL=T`.
2. **Timestep order.** In `internal_step.F:1092--1137`, `ADVANCE_SED` is called before `ADVANCE_PLAST`. The earlier offline-style block in `internal_step.F:213--235` has the same ordering. Plastic therefore sees the post-sediment state each internal step.

3. **Sediment dynamics.** `mod_sed.F:859--1156` advances active-layer thickness, settling/deposition, erosion, bed stratigraphy/bottom properties, horizontal advection, FCT limiting, vertical diffusion, OBC/source boundary conditions, nonnegative/upper clipping, and sediment exchange.
4. **Plastic dynamics.** `mod_plast.F:914--1851` computes ambient density and floc rate fields, updates plastic active-layer quantities, deposition/erosion/update-bottom, horizontal advection, vertical diffusion, kinetics, OBC/source boundary conditions, clipping, MPI exchange, and final recombination.
5. **Floc coupling to sediment.** In `mod_plast.F:4619--4815`, the aggregation rate calculation reads `sed(ised)%conc`, `sed(ised)%Sd50`, and `sed(ised)%Wset`. Thus active floc interaction is explicitly tied to sediment size and suspended sediment concentration. In case **b**, the saved output still has `mp1_agg=0`, so the current **b** run is effectively a split-disaggregated/null-floc test despite executing the floc branch.

Diagnosis

1. **Sediment runtime changed domain retention.** The drop from 1001.25 kg in `v001` to 582.41 kg in `v002` shows that turning on the sediment model changes the transport environment and/or open-boundary export enough to strongly affect domain MP1 mass. This is not necessarily a plastic mass sink by itself; the plotted domain mass does not include cumulative open-boundary export.
2. **BBL roughness is now demoted.** The `generic_sediment_a_no_bbl.inp` test suggests that changing the sediment BBL switches is not enough to recover the old case-a mass. Bottom-stress feedback should remain documented, but it is no longer the leading explanation for the present discrepancy.
3. **The no-floc plastic path itself changes when sediment is compiled/enabled.** In `mod_plast.F`, the update enters a `#if defined(SEDIMENT)` block before checking `floc_intract`. Even with `PLAST_FLOC=F`, case **a** uses the sediment-build no-floc branch around `mod_plast.F:1248--1278`. Therefore an old case-a run built without `-DSEDIMENT` is not a strict code-path match to a sediment-build case-a run.
4. **River forcing/source updating is a high-priority suspect.** The river update call changes at compile time. With plastic only, `bcond_gcn.F:645` calls `UPDATE_RIVERS` with `PLAST=PLASTDIS`; with sediment and plastic, `bcond_gcn.F:648` calls it with both `SED=SEDDIS` and `PLAST=PLASTDIS`. The plastic variables are still read in `mod_force.F:13096--13115`, but this optional-argument path should be verified directly by logging the instantaneous and cumulative river MP input, approximately $\sum_r Q_r C_{\text{plast},r} \Delta t$.
5. **ADVANCE_SED side effects remain possible.** `ADVANCE_SED` is called before `ADVANCE_PLAST` each internal step. With `MORPHO_MODEL=F`, bathymetry should not be updated, and with BBL disabled the roughness route is weakened. The next isolation is to keep `SEDIMENT_MODEL=T` but set `SED_START` to a very large value so sediment is initialized but `ADVANCE_SED` returns before doing transport. If plastic mass still follows the sediment-on result, the cause is setup/forcing/compile-path; if it returns toward the old result, the cause is inside `ADVANCE_SED` or its side effects.

6. **The remaining case-a/case-b gap is still a split-path problem after sediment state is matched.** With `SEDIMENT_MODEL=T` in both cases, the remaining 210.56 kg deficit in case b points back to differences between combined `conc` and split `conc_d` operators.
7. **The leading code-level suspects are water-column transport and boundary handling.** With cap counts zero and no sampled bed storage in case b, the likely divergence occurs during split `adv_scal`, split `vdif_scal`, split `bcond_scal_OBC`, split `bcond_scal_PTsource`, or nonnegative clipping before output.
8. **The split branch still has one awkward exchange point.** Before OBC handling, the floc branch exchanges `cnew_a`, `cnew_d`, and combined `cnew` (`mod_plast.F:1513--1516`). At that point the meaningful prognostic state is split; combined `cnew` is not rebuilt until `mod_plast.F:1715--1716`. This may be harmless, but it is not strictly symmetric with the no-floc path.
9. **Final recombination improved the budget but did not solve it.** The patched final order now clips and exchanges `cnew_a/cnew_d`, then rebuilds `conc = cnew = conc_a+conc_d`. The mass recovery after that change confirms it addressed a real asymmetry, but the remaining deficit shows another divergence exists upstream.
10. **Direct floc-rate reads from sediment should be bypassed in a true null-kinetics test.** The aggregation/disaggregation routines read `sed(ised)%conc`, `Sd50`, and `Wset`. For a strict “kinetics off” symmetry test, $\lambda_a = \lambda_d = 0$ should be imposed without calling sediment-dependent rate calculations at all. That removes any possible sediment-size or sediment-concentration dependence from case b.
11. **Case c is not a pure conservation comparator.** Case c retains much more total mass than case b, but it has active split components and bed storage. It is useful for full floc behavior, while case b remains the sharper test for whether split `conc_d` can reproduce combined `conc`.

Focused Next Checks

1. Use v002 as the new baseline for future plots and memo updates. The old v001 comparison should only be used to document the sediment-runtime effect.
2. Run a sediment-initialized but sediment-inactive case-a diagnostic by setting `SED_START` to a value beyond the run length. This separates sediment setup/forcing/compile effects from the runtime `ADVANCE_SED` operator.
3. Add a river-source budget log for MP1. At each report interval, record instantaneous $\sum_r Q_r C_{\text{plast},r}$ and cumulative $\sum_t \sum_r Q_r C_{\text{plast},r} \Delta t$. This will determine whether the apparent domain-mass loss is actually lower source input or stronger export.
4. Add a short internal checkpoint diagnostic for one small matched run. Record domain-integrated mass for `conc`, `conc_d`, and `conc_a+conc_d` after: start, deposition, erosion/update bottom, `adv_scal`, `vdif_scal`, OBC, point-source BC, nonnegative clamp, MPI exchange, and recombination.
5. For the null-floc case, add a temporary switch or guard so $\lambda_a = \lambda_d = 0$ is assigned directly and the sediment-dependent floc rate routines are not called.

6. Explicitly accumulate mass removed by nonnegative clipping. The hard cap should remain zero; any nonzero clipping loss would be immediately informative.
7. If serial and parallel runs differ, focus on the two exchange locations: `mod_plast.F:1513--1516` before OBC and `mod_plast.F:1677--1716` at final recombination.